

BASIC 3D PRINTING GUIDE FOR CREALITY ENDER-3 S1

3D printing can be a daunting experience when doing it for the first time. Fortunately, there are many resources available.

<https://www.creality3dofficial.com/blogs/news/ultimate-3d-printing-beginners-guide>

<https://ultimaker.com/learn/how-to-use-a-3d-printer/>

<https://all3dp.com/2/3d-printing-for-beginners-all-you-need-to-know-to-get-started/>

Installation

A 3D-printing utility is required. Install Ultimaker Cura: <https://ultimaker.com/software/ultimaker-cura/>

If you wish to create your own 3D models, use Autodesk Fusion. It is free to use:

<https://www.autodesk.com/products/fusion-360/overview>

Configuring the 3D printer in Cura

After installing Cura, configure the printer. Cura has a built in function to select a printer of known brands. Make sure to select Creality Ender-3 S1. Here is a very helpful beginners guide guide:

<https://all3dp.com/2/ender-3-s1-cura-profile-settings/#:~:text=Once installed%2C open Cura.,open the drop-down menu.>

Bed levelling

The preheat temperature is dependent on the type of filament. For any spool, the manufacturer usually indicates the recommended printing temperatures for the filament. You could use the center value in this range, or as a rule of thumb, use the following values:

PLA

- **Nozzle temperature:** 210 °C
- **Bed temperature:** 60 °C

ABS

- Nozzle temperature:** 235 °C

- Bed temperature:** 100 °C

PETG

- Nozzle temperature:** 240 °C

- Bed temperature:** 70 °C

To control the temperatures, on the Ender-3 S1 UI, go to **Control** → **Temperatures** → **Nozzle temperature/ Bed temperature** and change the nozzle temperature AND the bed temperature.

Above the extruder is a lever. When the filament is preheated, press the lever to the side to release the filament, and pull the filament from the extruder (from the top).

Now the printer is ready for bed leveling. Using the steel brush in the drawer of the S1 under the UI, clean off any remaining filament from the nozzle. Be careful not to snag any of the wires. Good practice is to use a nozzle cleaning needle (0.2mm for 0.2mm nozzle) by pressing the extruder lever and doing a few passes of the needle all the way through from the filament entry point to the nozzle, cleaning the tip of the needle from any caught filament as you go. If the tip is clean, and everything is heated to the desired temperatures, bed leveling can be done.

These videos are good resources for bed levelling, however not all preheat the nozzle or bed before leveling. This step is very important for clean prints. The rest (aux. Leveling with paper, auto leveling, storing configuration) can be followed as done in the videos:

<https://www.youtube.com/watch?v=3VM5RK4nF94>

<https://www.youtube.com/watch?v=raSW7cDGOBg&t=1336s>

Bed leveling does not have to be done with every print. Every 10-15 prints should be fine.

First things to do when loading a new filament

Good practice for loading new filament is to print a temperature tower. See:

<https://all3dp.com/2/temp-tower-cura-tutorial/>

Based on the temperature tower you can determine what the optimal nozzle temperature will be for that type of filament, on that spool. This is not as crucial for PLA, but printing temperature towers for trickier filaments such as PETG reduces waste and time as it reduces the risk of failed prints.

Configuring a model for printing

In Cura, the .STL file of a 3D model can be dragged onto the build plate. The model can be moved, scaled, rotated. What's most important is setting the temperatures to match the filament used, ensuring that the correct face of the object is on the build plate, orientation and knowing when to use supports.

Orientation: <https://all3dp.com/2/3d-print-orientation-troubleshooting/>

Supports: <https://all3dp.com/1/3d-printing-support-structures/>

Build plate adhesion: <https://all3dp.com/2/3d-printer-bed-adhesion-all-you-need-to-know/>

Where to find 3D models

More often than not, if you think of anything that you'd like to print, someone has already made it, printed it, and documented it. Websites such as Thingiverse.com or Printables.com are resources for free, open source 3D models of basically anything.

<https://www.thingiverse.com/>

<https://www.printables.com/>

Some tips

- Do not print supports unless there are overhangs on the print.
- Always preheat before printing!
- Prints that have a large surface area sometimes require a build plate adhesion, such as brim to avoid warping.
- Playing around with the printer is the best way to learn how to use it.
- You will have failed prints, and there are various reasons why. Work through it gradually, from adjusting temperatures slightly, redoing bed leveling, or even tweaking while printing. For example, if you have leveled the bed, but you see the filament is not sticking, you can adjust the z-height, bed and nozzle temperatures, print speeds and z-height. Just remember that if you change the temperature during a print, then cancel/finish the print, the temperatures of the FILE will be loaded the next time that object prints, not the temperature you set during the previous print. These temps must be adjusted in the print utility (Cura).
- Have fun!